Faculty of Engineering Technology
Basic Science Department
Department of Physics and Basic Sciences

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية
قسم الفيزياء والعلوم الأساسية

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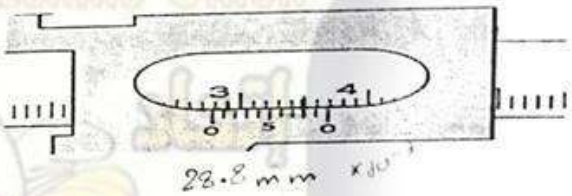
Midterm Exam Phy-111(C)

الاسم بالعربية : ر ... ن المادة ج١ ... موعد المختبر : ث (١١ - ٢)

$4 \times \frac{1}{2}$ $\frac{1}{2} \times \frac{1}{3}$

- *Q 1: Fill the blank with suitable answer :* 4.5 x 1 = 4.5 *(13.5 marks)*

1) If we turned a closed micrometer (9) turns and (1/5) of a turn, then the micrometer reading is ( 7.5 mm).




28.8 mm x10⁻³

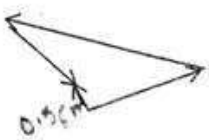
2) L= ( 0.0288 m) ± ΔL=( 0.05m) .




D = 2.12
r = 1.06 mm

3) If the diameter for a ball was measured using a micrometer as shown then: $\Delta A$ = ( 1.41 x10⁻⁶ $m^2$).

4 ) If the scale used to draw this vector diagram for three forces was ( 2mm = 3 N ),then the magnitude of the net force is( 7.5 )




5) The instrument used to take this reading (11.35 cm) is (Verniercaliper,Micrometer,Ruler).

( Vernier Caliper ).

6) Using significant rules to find 37600/4. = ( .9 X10+4 ).

7) In gravitational potential energy experiment, before start move toward down from ( $h_i$=70cm) the mass was (m=20g) and ( M=200g) ,then the kinetic energy when (m) reach ( h=40cm) is: ( 0.058 joule).

8) For the above question the final velocity ($V_f$) when m reach ground is: ( 1.146 m/s ).

9) Using significant rules to find , 6.299+3.78= ( 10.08 ).

• *Q2 : Circle the correct answer:( 10.5 marks )*

1) If a student used a ruler with a maximum range (20 cm) to measure the length of a table which has the length (1.7 m), then the uncertainty in length (ΔL) is:

a) 3mm b) 4.5mm c) 8.5mm d) 6mm

2) If the relation between A&B given as (C =A-B), if A = 5.28 and B=1.8,then (C±ΔC) is:

(Using significant rules)

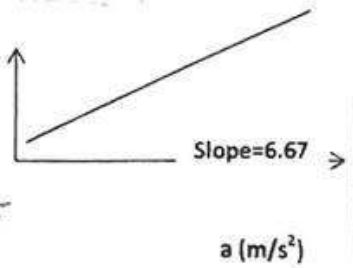
a) 3.48±0.05 b) 3.5±0.1 c) 3.5±0.06 d) 3.48±0.055

3) F1,F2,F3 are three forces in equilibrium ,if you know that ( $F_{3x}$ =1.32 N , $F_{3y}$ =7.5 N ),the direction of resultant between (F1,& F2) according to x-positive axis is :

a) 100 b) 80 c) 280 d) 260

4) In Newton's second law experiment, if we got this graph,

then the constant hanged mass in the experiment is:




a) 15g b) 25g

c) 60g d) 80g

slop = 1/F

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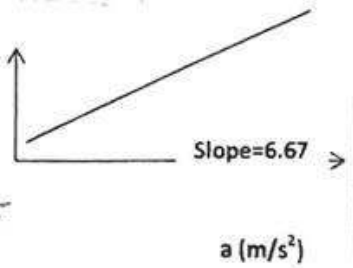
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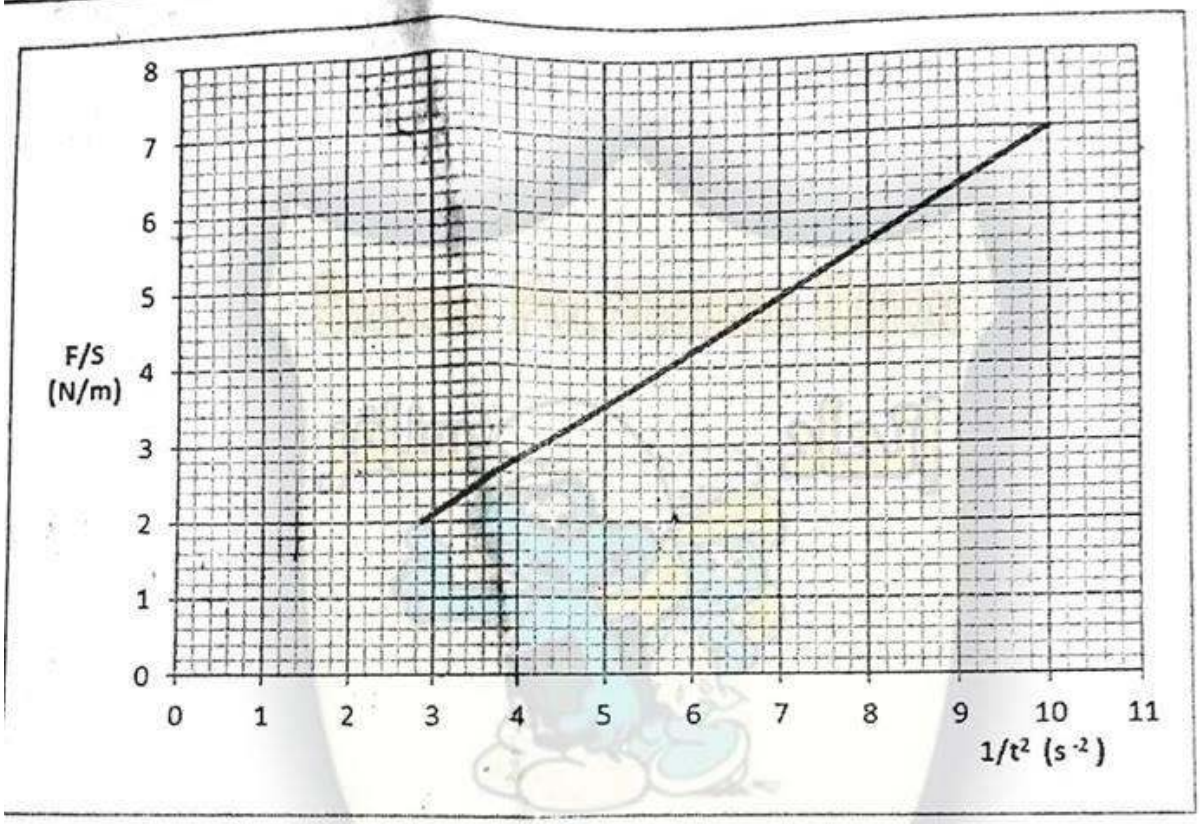
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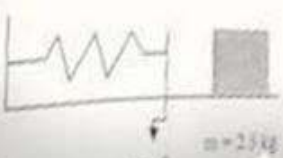
slop = 1/F


F/S
(N/m)
0
1
2
3
4
5
6
7
8
0
1
2
3
4
5
6
7
8
9
10
11
1/t² (s⁻²)

13. Suppose that the work done by an engine is given by $W = 7t^2 + 40t + 100$, where the units are SI. The power (in watt) developed by this engine at $t = 2$ s is:

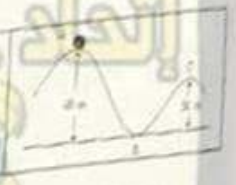
a) 68 b) 82 c) 96 d) 54

14. A mass m = 2.5 kg is sliding along a frictionless table with initial speed v as shown in the figure. It strikes a spring that has a force constant k = 1000 N/m and compresses it from its equilibrium position a distance 5 cm ($x_f$ = -5cm). The initial speed v of the block was (in m/s):




a) 1.22 b) 1.41 c) 0.71 d) 1

15. A sphere of mass m slides on a frictionless path ABC. If the sphere starts from rest at point A, the speed of the sphere (in m/s) at point C is:



a) 24.5 b) 17.3 c) 14.1 d) 31.6

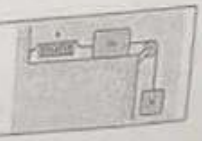
16. The potential energy function for a system is given by $U(x) = -x^3 - 2x^2 + 3x$. The magnitude of the force (in N) at x = 3 m is:

a) 92 N b) 17 N c) 36 N d) 61 N

17. A particle (m = 4 kg) moves along the x axis. Its position varies with time according to $x(t) = t^2 + 2t$, where x is in meters and t is in seconds. The kinetic energy of the particle at time t = 3 s:

a) 200 J b) 128 J c) 72 J d) 32 J

18. The blocks shown are released from rest with the spring unstretched. The pulley and the horizontal surface are frictionless. If k = 150 N/m and M = 1 kg, what is the maximum extension of the spring?



a) 13.3 cm b) 26.7 cm c) 20 cm d) 6.7 cm

6. A spring has compressed a distance x = 0.1 m. If the spring constant k = 200 N/m, calculate the potential energy stored in the spring?

a) 1 J  b) 2.25 J  c) 4 J  d) 0.25 J

7. A force F (F = 40 N), parallel to the incline, is required to push a certain object at constant velocity up a frictionless incline that is 30° above the horizontal. The mass of the object is:



a) 6.4 kg  b) 5.8 kg  c) 10.4 kg  d) 8.0 kg

8. 2000 kg car - moving at an initial speed of 25 m/s along a horizontal road - can brake to stop in a distance 80 m. The coefficient of kinetic friction between the tires and the road is:

a) 0.26  b) 0.39  c) 0.52  d) 0.31

9. When an object is moving in a circle at constant speed, its acceleration is

a. constantly increasing.
b. constant in direction.
c. zero.
d. constant in magnitude.

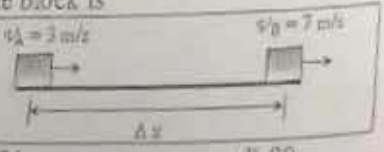
10. A racing car rounds a curve at a speed of 30 m/s. If the force needed to provide the centripetal acceleration is equal to the weight of the car, the radius of the curve is:

a) 90 m  b) 160 m  c) 250 m  d) 40 m

11. The coefficient of static friction between a 6 kg-block and horizontal surface is 0.4. What is the maximum horizontal force that can be applied to the block before it moves?

a) 32 N  b) 20 N  c) 24 N  d) 28 N

12. A block of mass = 4 kg has a speed of 3 m/s initially (at point A) and a speed of 7 m/s at point B. Then the total work (in J) done on the block is




a) 154  b) 108  c) 21  d) 80